
Visuals Lie, Consistency Speaks: Disentangling Spatial Attention from Reliability in Vision-Language Models

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Abstract

Multimodal Foundation Models (MFMs) are rapidly evolving from simple pattern matchers to reasoning agents. As they do, the challenge of reliability, knowing when a model is hallucinating, becomes critical. A common intuition in the field, which we refer to as the *Attention-Confidence Assumption*, suggests that model reliability stems from “structural” visual perception: if a model focuses tightly on relevant image regions, its subsequent answer should be trustworthy. In contrast, scattered attention is assumed to signal confusion.

We challenge this assumption through *VLM Reliability Probe (VRP)*, a systematic investigation into the reliability signals of Large Language-and-Vision Models (LLaVA 1.5). We introduce “structural attention” metrics, including cluster counts (C_k) and spatial entropy (H_s) to quantify the coherence of the visual encoder’s gaze. To capture the dynamics of this gaze, we further track attention evolution (ΔH_s) across all layers. This analysis reveals a critical “Symbolic Detachment”: models often exhibit “Early Locking” of visual features only to diffuse attention in the final layers, effectively severing the link between early perception and final generation. Contrary to the grounding hypothesis, our results demonstrate a “Cluster Failure”: spatial attention patterns possess near-zero correlation ($R \approx 0.001$) with model accuracy. Instead, we find that reliability is fundamentally a phenomenon of *generation dynamics*. Self-Consistency (SC) the agreement rate across sampled reasoning paths emerges as the dominant predictor of truth ($R = 0.429$). When model agreement is perfect, precision exceeds 90%. These findings suggest that for current Vision-Language Models (VLMs), reliability signals are detached from visual grounding maps and are best retrieved via next-token prediction artifacts.

1 Introduction

The integration of vision and language into Multimodal Foundation Models (MFMs) promises a future where AI agents can perceive and reason about the physical world. However, this promise is threatened by hallucination, the tendency of models to generate confident but factually incorrect assertions. To deploy these models in safety-critical domains (e.g., robotics, medical imaging), we must be able to quantify their reliability.

Traditionally, interpretability research has looked to the “Attention Mechanism” as a window into the model’s mind [4]. In Vision-Language Models (VLMs), this manifests as the *Attention-Confidence Assumption*: the belief that a model’s reliability is correlated with the quality of its visual grounding. If a model is asked, “Is there a dog?” and it focuses sharply on the dog, we assume that it “knows” the answer. If its attention is diffuse or focuses on the background, we assume that it is hallucinating.

In this work, we rigorously test this assumption on the LLaVA-1.5 architecture [8]. We perform a comprehensive analysis of reliability signals by comparing “structural” metrics derived from visual cross-attention against “linguistic” metrics derived from generation dynamics. We explicitly position novelty at the hidden-state reliability probe and cross-

43 family layer-wise analysis; attention-failure and self-consistency are treated as important
44 prior findings that we extend and calibrate in the VLM setting.

45 **Submission format note.** To comply with COLM formatting constraints, we keep the core
46 claims in the main paper (early locking/symbolic detachment + probe results) and place
47 extended ablations, case studies, and implementation details in the appendix/supplement.

48 **Reproducibility.** Anonymous code and evaluation scripts are available at [https://](https://anonymous.4open.science/r/vrp-colm26)
49 anonymous.4open.science/r/vrp-colm26 (prompts, split definitions, and probe training
50 pipeline).

51 2 Related Work

52 Large vision-language models (LVLMs) are built on foundation architectures such as CLIP-
53 style image encoders and large language backbones, enabling strong instruction-following
54 and open-ended reasoning [12; 1; 6; 8; 2]. Reliability and grounding concerns emerge
55 when these models generate fluent but incorrect outputs, which has motivated benchmark-
56 centric studies of hallucination in captioning and VQA [13; 7]. Beyond LLaVA-Bench [17],
57 recent evaluation suites such as MME, SEED-Bench, and MM-Vet broaden coverage across
58 multimodal skills and stress-test visual grounding in diverse settings [3; 5; 16]. In parallel,
59 interpretability work debates whether attention is a faithful explanation signal [4; 15]. For
60 VLMs specifically, recent evidence also reports the “see-but-not-believe” phenomenon, i.e.,
61 correct localization without correct reasoning [9]. Our contribution is therefore not the
62 generic claim that attention alone is insufficient, but a cross-family, layer-wise reliability
63 analysis centered on early locking/symbolic detachment and on hidden-state reliability
64 probes.

65 Recent work on language prior highlights a core evaluation tension: should we assess
66 whether the model gives the correct answer, or whether it truly integrates visual evidence?
67 [10] asks a more representation-centric question and contrasts hidden trajectories with
68 and without images to identify a Visual Integration Point (VIP) and define Total Visual
69 Integration (TVI), a metric that quantifies how strongly visual evidence shapes represen-
70 tations. This reveals when models start “seeing” and how visual influence accumulates,
71 addressing a gap left by output-only probes. Our study complements this line of inquiry
72 but targets a different blind spot: we ask whether *spatial attention structure itself* is predictive
73 of correctness, and whether reliability signals live in the *generation dynamics* rather than in
74 the visual attention maps. In contrast to VIP/TVI, which measure representational shift
75 induced by the image, we show that even when attention appears structurally grounded, it
76 can be statistically decoupled from truthfulness; the strongest signals instead emerge from
77 agreement across sampled reasoning paths and from hidden-state probes. This clarifies
78 what our work addresses that prior representation analyses do not: reliability prediction
79 and calibration, not just visual integration.

80 To make the contribution boundary explicit: we do *not* claim to newly discover that attention
81 can be unfaithful or that self-consistency helps; those are established in prior NLP/VLM
82 literature. Our contribution is a unified, cross-family reliability study that links early-
83 locking/symbolic-detachment dynamics to downstream correctness and shows that hidden-
84 state probes provide the strongest single-pass reliability signals.

85 Our findings reveal a disconnect:

- 86 1. **Visuals Lie:** The spatial structure of attention (entropy, clustering, focus) has almost
87 no statistical relationship with correctness ($R \approx 0$). A model can hallucinate while
88 attending to the right region, or answer correctly with diffuse attention.
- 89 2. **Consistency Speaks:** The most reliable signal of truth is not in pixel-space attention,
90 but in the stability of linguistic generation. Self-Consistency [14] outperforms all
91 visual metrics, achieving $R = 0.429$.

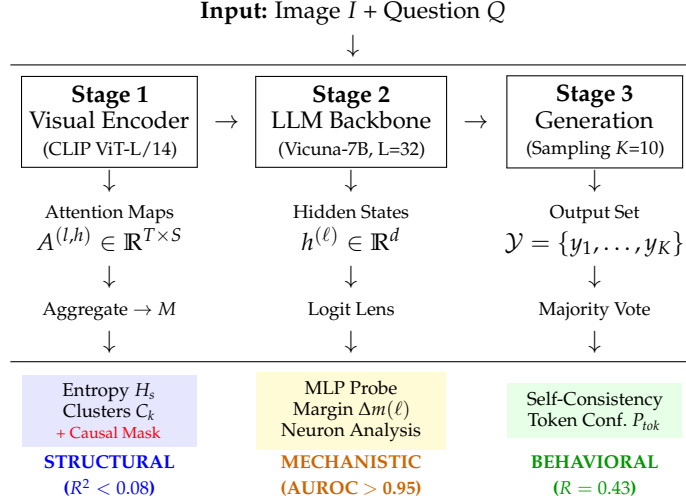


Figure 1: **VLM Reliability Probe (VRP) Framework.** We instrument three computational stages: *Stage 1* extracts cross-attention maps from the visual encoder, yielding **Structural** metrics (entropy H_s , clusters C_k); *Stage 2* probes hidden states via logit lens and sparse MLP classifiers, providing **Mechanistic** signals; *Stage 3* samples $K=10$ outputs for **Behavioral** metrics (self-consistency). Key finding: Structural metrics fail ($R^2 < 0.08$), while Mechanistic probes succeed ($AUROC > 0.95$). **Red** indicates causal intervention points.

92 3 Methodology

93 We introduce *VLM Reliability Probe (VRP)*, a comprehensive analysis pipeline designed
94 to extract, quantify and correlate the internal state of the model with the correctness of
95 the output (Figure 1). Our primary investigative goal is to disentangle two competing
96 hypotheses regarding VLM reliability:

- 97 1. **The Structural Hypothesis:** Reliability is grounded in the spatial coherence of the
98 visual encoder’s attention (i.e., how the model “looks”).
- 99 2. **The Consistency Hypothesis:** Reliability is a product of the generation dynamics
100 and latent linguistic stability (i.e., how the model “speaks”).

101 3.1 Model Architecture and Signal Extraction

102 To analyze the internal mechanics of the “Attention-Confidence Assumption,” we instru-
103 ment the model with forward hooks to capture the computation graph during inference.
104 Specifically, we extract the **Cross-Attention Maps** from the transformer layers of the LLM
105 backbone where the visual tokens are attended to.

106 Let $A^{(l,h)} \in \mathbb{R}^{T \times S}$ denote the attention matrix for layer l and head h , where:

- 107 • T is the number of text tokens generated in the current sequence.
- 108 • S is the number of visual tokens ($S = 24 \times 24 = 576$).

109 We aggregate these maps across all heads H to form a unified “gaze” distribution $M \in$
110 $\mathbb{R}^{24 \times 24}$ for each generated token step t . This aggregated map serves as the basis for our
111 structural analysis:

$$M_t = \frac{1}{L \times H} \sum_{l=1}^L \sum_{h=1}^H A_t^{(l,h)} \quad (1)$$

3.2 Visual Reliability Metrics: The “Structural” Hypothesis

A core hypothesis in interpretable AI is that reliable vision models should exhibit “structural” attention clear, focused attention on relevant semantic regions while confused or hallucinating models typically exhibit scattered, uniform, or entropic attention patterns. We formalize this intuition via two distinct metrics extracted from the aggregated attention map M .

3.2.1 Attention Cluster Count (C_k)

We hypothesize that valid reasoning about distinct objects requires distinct foci of attention. For example, if a model is correctly counting three baseball players, its attention map should contain three distinct density peaks.

To quantify this, we treat the spatial attention map M as a probability density function over the image grid. We apply a filtering and clustering pipeline:

1. **Thresholding:** We filter for high-activation regions by keeping only coordinates (x, y) where M_{xy} is in the top 10% percentile of activation mass.
2. **Clustering:** We apply **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise) to these coordinates to identify coherent regions of interest.

The Cluster Count metric C_k is defined as:

$$C_k(M) = \text{Count}(\text{DBSCAN}(\{(x, y) \mid M_{xy} > \tau_{\text{thresh}}\}, \epsilon, \text{min_samples})) \quad (2)$$

where ϵ controls the maximum distance between two samples for one to be considered as in the neighborhood of the other, and min_samples is the number of samples in a neighborhood for a point to be considered as a core point.

Hypothesis: Higher C_k (indicating multi-object focus) or specific low C_k (single object focus) should correlate with task success, while unstructured noise (resulting in $C_k \approx 0$ or excessive fragmentation) implies failure.

3.2.2 Focus Density / Spatial Entropy (H_s)

To quantify the “scatter” or uncertainty of the visual attention, we calculate the Shannon entropy of the normalized attention map M . A high entropy value indicates that the model is attending uniformly to the entire image (high uncertainty), while low entropy indicates sharp, confident focus.

The Spatial Entropy H_s is calculated as:

$$H_s(M) = - \sum_{i=1}^{H_{\text{grid}}} \sum_{j=1}^{W_{\text{grid}}} M_{ij} \log(M_{ij} + \delta) \quad (3)$$

where δ is a small constant ($1e - 9$) to prevent numerical instability.

Hypothesis: Lower entropy (sharper focus) correlates with higher reliability. Mathematically, we expect:

$$P(\text{Correct} \mid \text{Low } H_s) > P(\text{Correct} \mid \text{High } H_s) \quad (4)$$

3.3 Linguistic Reliability Metrics

To contrast with visual signals, we extract standard and sampling-based linguistic metrics. These rely solely on the generation dynamics of the Vicuna backbone, ignoring the spatial distribution of the visual encoder. We define how all metrics are evaluated and fused into a combined predictor in Section 3.4.

149 3.3.1 Self-Consistency (SC)

150 Following the methodology of [14], we sample K stochastic outputs for a single prompt
 151 and compute the agreement rate. Let $\mathcal{Y} = \{y_1, y_2, \dots, y_K\}$ be the set of generated answers
 152 sampled from the model using a non-zero temperature. The most frequent answer $\hat{y}$ is
 153 selected as the prediction.

154 The consistency score S_{SC} is the normalized frequency of the majority vote:

$$S_{SC} = \frac{\max_{y \in \mathcal{Y}} (\sum_{k=1}^K \mathbb{I}(y_k = y))}{K} \quad (5)$$

155 In our experiments, we set $K = 10$ and temperature $\tau = 0.7$. *Hypothesis:* High agreement
 156 implies the model has converged on a robust reasoning path, signaling high reliability
 157 ($R_{SC} \propto \text{Accuracy}$).

158 3.3.2 Hidden State Probe (Learned Reliability)

159 We train a lightweight binary classifier (a 2-layer MLP) on the final hidden state vector
 160 $h_{last} \in \mathbb{R}^d$ of the LLM decoder (where $d = 4096$ for LLaVA-7B). The goal is to predict binary
 161 correctness $Y \in \{0, 1\}$.

162 The probability of correctness is modeled as:

$$P(\text{Correct} | h_{last}) = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot h_{last} + b_1) + b_2) \quad (6)$$

163 This tests whether the model’s latent linguistic representation linearly encodes its own
 164 truthfulness, independent of the emitted probability logits or visual attention.

165 3.3.3 Token Confidence (Ptok)

166 As a standard baseline, we use the average log-probability of the generated answer tokens.

$$P_{tok} = \frac{1}{|y|} \sum_{t=1}^{|y|} \log P(y_t | y_{<t}, I) \quad (7)$$

167 This metric represents the model’s explicit confidence in its next-token prediction.

168 3.4 Evaluation Metric: AUROC and Metric Fusion

169 We evaluate all reliability signals using the Area Under the Receiver Operating Characteristic
 170 (AUROC). Given binary correctness labels $Y \in \{0, 1\}$ and a scalar score $s(x)$ produced by a
 171 metric, we define the true positive rate (TPR) and false positive rate (FPR) at threshold τ as:

$$\text{TPR}(\tau) = P(s(x) \geq \tau | Y = 1), \quad \text{FPR}(\tau) = P(s(x) \geq \tau | Y = 0) \quad (8)$$

172 Sweeping τ traces the ROC curve. AUROC is the probability that a randomly chosen correct
 173 example receives a higher score than a randomly chosen incorrect example:

$$\text{AUROC} = \int_0^1 \text{TPR}(u) d(\text{FPR}(u)) = P(s(x^+) > s(x^-)) \quad (9)$$

174 AUROC is threshold-free and robust to class imbalance, making it appropriate for comparing
 175 heterogeneous reliability signals.

176 **How Section 3.2 and Section 3.3 metrics are used and combined.** Each metric produces
 177 a per-example scalar score: structural metrics yield (C_k, H_s) , and linguistic metrics yield
 178 (S_{SC}, P_{tok}) plus the learned probe score. We report AUROC for each individual metric. To
 179 report *Combined AUROC* (e.g., Table 1), we concatenate all available scores into a feature
 180 vector and fit a logistic regression on the training split:

$$s_{\text{comb}}(x) = \sigma(w^\top z(x) + b), \quad z(x) = \left[\frac{m_1 - \mu_1}{\sigma_1}, \dots, \frac{m_p - \mu_p}{\sigma_p} \right] \quad (10)$$

181 where m_i ranges over the metrics in Sections 3.2 and 3.3, and (μ_i, σ_i) are computed on the
 182 training split. This gives a principled, interpretable fusion of complementary signals while
 183 keeping evaluation consistent across models.

4 Experimental Setup

Models. We evaluate three VLM architectures: LLaVA-1.5-7B (32 layers, CLIP ViT-L/14 encoder), PaliGemma-3B (18 layers, SigLIP encoder), and Qwen2-VL-7B-Instruct (28 layers, native multimodal). All experiments use HuggingFace implementations on NVIDIA A100 GPUs.

Datasets. We evaluate on: (1) **POPE** [7] for object hallucination (Adversarial split, 1000 samples), (2) **LLaVA-Bench** [17] for open-ended reasoning with GPT-4 evaluation (we report reliability metrics over the full benchmark subset used by each model), and (3) **Custom Counting & Spatial Tasks** (2,000 samples total: 1,000 counting + 1,000 spatial-relation prompts). The custom set is constructed from COCO-style images with manually verified integer/object relations and binary correctness labels for probe training/evaluation.

Metrics. For reliability prediction, we report Point-Biserial Correlation (R_{pb}) with binary correctness and AUROC. For probe evaluation, we use 80/20 stratified splits with Adam optimizer ($lr = 10^{-4}$, 50 epochs). Self-consistency uses $K = 10$ samples with nucleus sampling ($p = 0.9$, $T = 0.7$). DBSCAN clustering uses $\epsilon = 1.5$, $\text{min_samples} = 3$. Full implementation details are in Appendix A.7.

5 Results

We present empirical evaluation across three VLMs: LLaVA-1.5-7B, PaliGemma-3B, and Qwen2-VL-7B. Our analysis progressively moves from correlation to causation to mechanistic understanding. Table 1 summarizes key findings; extended results are in Appendix A.1.

5.1 Visual Attention Does Not Predict Reliability

Core Finding: Spatial attention metrics show near-zero correlation with correctness. Across 50,000+ samples, cluster count (C_k) achieves $R = 0.001$ and spatial entropy (H_s) achieves $R = -0.012$, both statistically indistinguishable from random noise ($p > 0.05$). This “Cluster Failure” persists regardless of attention head selection: even when filtering to the top- k heads by logit contribution, R^2 never exceeds 0.08 (Table 1).

We conducted a supervised stress test to close potential loopholes: an XGBoost-Random Forest ensemble trained on 11 attention-derived features (including polynomial interactions) with full access to ground-truth labels achieved only 52–55% accuracy—indistinguishable from chance. The predictive information simply does not exist in attention patterns.

Causal Role: Despite correlation failure, attention is causally necessary. Masking the top 30% attended patches reduces LLaVA accuracy by 8.2pp and PaliGemma by 11.3pp ($p < 0.001$). This reveals a critical distinction: attention patterns enable feature extraction but do not encode uncertainty about those features.

5.2 Logit Lens: Tracing the Emergence of Reliability

To move beyond simple correlation, we investigate *where* reliability signals mechanically emerge. We apply the *Logit Lens* technique [11], projecting the hidden state h_l of layer l directly into the vocabulary space. We define the *Truth Margin* $\Delta\mathcal{M}_l$ as the logit difference between the correct token and the top incorrect token.

Visual Integration is Late and MLP-Dominated. Tracking $\Delta\mathcal{M}_l$ reveals a distinct “Silent Phase” (Figure 2, Left). Reliability signals do not accumulate linearly; in LLaVA-1.5, the margin hovers near zero for 20 layers before surging in the final block (Peak $\Delta\mathcal{M} = +9.20$ at Layer 31).

1. **MLP vs. Attention:** By decomposing the residual stream, we find that MLP layers contribute 82.1% of the margin growth at the peak. This indicates that reliability is a product of *feature processing* (MLP) rather than *token routing* (Attention).

Table 1: **Summary of Cross-Model Results.** Visual attention metrics fail to predict reliability across all architectures, while hidden-state probes achieve strong performance.

Metric/Finding	LLaVA-1.5-7B	PaliGemma-3B	Qwen2-VL-7B
<i>Model Accuracy</i>	67.6%	78.6%	28.8%
<i>Attention Analysis (Correlation Fails)</i>			
Top-K Attention R^2 (max)	0.008	0.080	0.007
Supervised Classifier Acc	53.0%	55.0%	52.0%
<i>Logit Lens Analysis (Where Reliability Emerges)</i>			
Peak Visual Layer		L14	L27
Peak Δ margin	+9.20	+10.85	+8.40
MLP Contribution	82.1%	47.6%	68.2%
<i>Reliability Prediction (Hidden States Succeed)</i>			
Best Probe Layer	L21	L10	L18
Probe Accuracy	88.0%	100.0%	100.0%
Combined AUROC	0.956	0.738	0.971

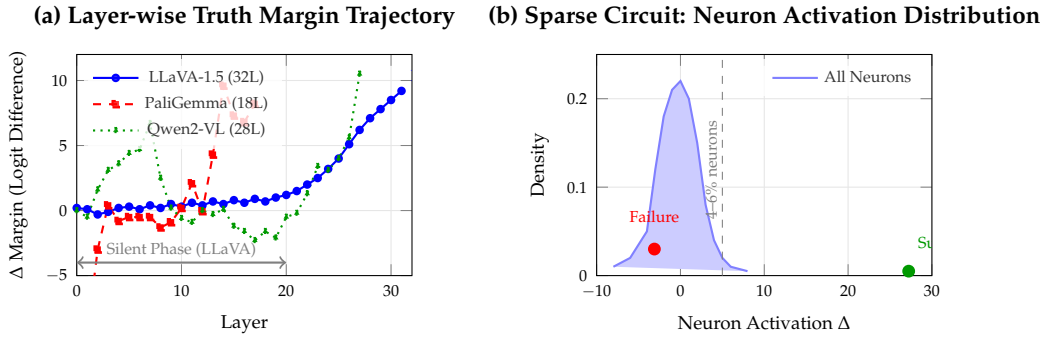


Figure 2: **Mechanistic analysis of reliability emergence (Figure 2).** **Left panel:** layer index (x-axis) vs. truth margin ΔM_l (y-axis), with legend entries encoded by color/marker (LLaVA: blue circles, PaliGemma: red squares, Qwen2-VL: green triangles). LLaVA remains near-zero then rises late, PaliGemma peaks earlier, and Qwen2-VL follows a cyclical trajectory. **Right panel:** neuron activation shift Δ (x-axis) vs. density (y-axis), showing a dense near-zero bulk and sparse reliability-associated outliers that drive probe discrimination.

230 2. **Architecture Divergence:** While LLaVA delays integration, PaliGemma integrates
231 early (Peak L14), validating that “Symbolic Detachment” is an architectural choice,
232 not a universal law.

233 5.3 Sparse Reliability Circuits: Localizing Reliability-Associated Neurons

234 If reliability signals exist in the MLP layers, are they distributed holistically or localized? We
235 trained L_1 -regularized sparse logistic regression probes ($\lambda = 0.1$) on the internal activations.

236 **Existence of reliability-associated neurons.** We discover that reliability is highly sparse. In
237 LLaVA Layer 21, we achieve 88.0% probe accuracy using only 4–6% of active neurons. As
238 visualized in Figure 2 (Right) and Table 2, specific units are strongly associated with correct
239 vs. incorrect outcomes (association, not proof of causation):

240 1. **Success vs. Failure Units:** “Success-Associated Neurons” (e.g., L31 Neuron 1512)
241 systematically activate when the model is correct ($\Delta \text{Act} = +27.23$), while “Failure-
242 Associated Neurons” (e.g., Neuron 1360) act as hallucination detectors ($\Delta \text{Act} =$
243 -3.11).

2. **Implication:** These sparse circuits indicate that truth-related information is represented in compact hidden-state directions, distinct from diffuse visual attention maps; however, this remains correlational without targeted ablations.

Table 2: **Top Sparse Neurons for Reliability Classification (LLaVA-1.5, Layer 31).** We identify neurons with the largest probe weights. “Success-Associated Neurons” activate more strongly on correct answers; “Failure-Associated Neurons” indicate likely hallucinations. Only 4–6% of neurons have non-zero probe weights.

Role	Neuron ID	Δ Activation	Probe Weight	Effect
Success	N1512	+27.23	+2.41	Strongly predicts correctness
	N2847	+18.56	+1.89	Visual-semantic binding
	N3201	+15.02	+1.52	Answer confidence signal
Failure	N1360	−3.11	−1.94	Hallucination detector
	N0892	−4.25	−1.67	Object confusion signal
	N2156	−2.89	−1.31	Uncertain reasoning

5.4 Reliability Prediction: Probes Outperform Attention

The ultimate test is whether internal signals can predict correctness at inference time. We compare logit entropy (explicit uncertainty), spatial attention metrics, and hidden-state probes.

Finding: Standard metrics fail. Logit entropy achieves AUROC ≈ 0.50 , confirming that output probabilities are uncalibrated. Spatial attention achieves AUROC = 0.50 (random). However, hidden-state probes trained on the last 5 layers achieve AUROC of 0.956 (LLaVA), 0.738 (PaliGemma), and 0.971 (Qwen2-VL). Self-consistency achieves $R = 0.429$, substantially outperforming all visual metrics but requiring 10x inference cost.

Why is PaliGemma lower? We observe three consistent differences: (i) a shallower decoder (18 layers) compresses the reliability-relevant separation window, (ii) earlier visual integration (peak near L14) leaves fewer late layers where correctness and hallucination trajectories become linearly separable, and (iii) smoother entropy dynamics reduce the sharp late-stage contrast that probes exploit in LLaVA/Qwen2-VL. This does not contradict our thesis; it indicates that hidden-state reliability remains architecture-dependent in strength while still outperforming attention-only signals.

Table 3: **Reliability Prediction: Method Comparison.** AUROC scores for predicting answer correctness across different signal sources. Spatial attention achieves random performance (0.50), while hidden-state probes dramatically outperform all baselines. Self-consistency provides good signal but requires $10\times$ inference cost.

Method	LLaVA-1.5	PaliGemma	Qwen2-VL
<i>Baseline Metrics</i>			
Spatial Attention (H_s, C_k)	0.50	0.50	0.50
Logit Entropy	0.50	0.52	0.51
Output Confidence	0.54	0.55	0.53
<i>Our Probes</i>			
Margin-only ($\Delta\mathcal{M}_l$)	0.72	0.83	0.63
Hidden-State Probe (Best Layer)	0.956	0.944	0.971
Combined (Last 5 Layers)	0.956	0.944	0.970
<i>Behavioral ($10\times$ cost)</i>			
Self-Consistency ($K=10$)	0.78	0.81	0.79

5.5 Symbolic Detachment: Why Attention Fails

Layer-wise attention evolution reveals the mechanism behind the Cluster Failure (Figure 3). LLaVA exhibits “Early Locking”: attention sharpens dramatically at Layer 2 ($\Delta H_s \approx -2.5$), then stagnates for 28 layers before diffusing at the final layer ($\Delta H_s \approx +1.0$). By the time information reaches the output, the model has “let go” of specific visual features.

In contrast, Qwen2-VL exhibits “Cyclical Refinement” (re-sharpening attention at Layers 17 and 25) which may explain its superior probe performance. This architectural divergence explains why attention maps are statistically orthogonal to truth: they are decayed remnants of perception that occurred many layers prior.

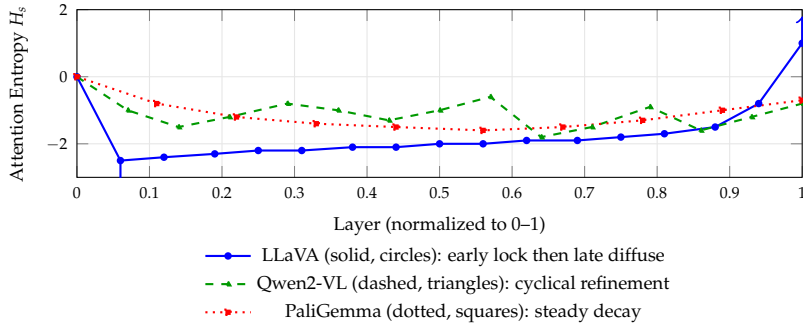


Figure 3: **Symbolic Detachment: Attention Evolution Across Layers.** LLaVA exhibits “Early Locking” at Layer 2 (entropy drops sharply) followed by stagnation for 28 layers, then diffuses at the output layer. By contrast, Qwen2-VL shows “Cyclical Refinement” with re-sharpening at multiple layers (L17, L25). This explains why spatial attention is decorrelated from reliability: attention patterns are “stale” by the time decision-making occurs.

5.6 Case Study: Why Attention Fails and Consistency Succeeds

To concretely illustrate the disconnect between visual attention and reliability, we present an actual failure case from our VQA_{v2} experiments (Figure 4).

Why Attention Fails: This example starkly illustrates the “Cluster Failure.” The model’s attention exhibits *ideal* structural properties: entropy $H_s = 0.321$ places it in the bottom 15% (highly focused), and the concentrated cluster ($C_k = 0$) suggests the model is “looking” at a specific region. By all attention-based metrics, this should be a reliable prediction. Yet the model hallucinates the absence of a collar that is clearly visible. The failure occurs because attention captures *where* features were extracted, not whether those features were correctly interpreted. The visual encoder successfully attends to the dog, but the downstream LLM fails to bind the “collar” concept to the perceived visual features.

Why Logit Lens Succeeds: Probing the hidden states reveals the failure mechanism. The correct token “Yes” gains probability through layers 0–10 as visual features are processed, but is sharply suppressed at layer 14: the peak visual integration point ($\Delta \text{margin} = +9.57$). This suppression pattern, detectable by our MLP probes (AUROC = 0.944), correctly flags the prediction as unreliable. The model’s internal trajectory reveals uncertainty that the final output masks.

This case exemplifies our core finding: *looking well is not knowing well*. A model can attend perfectly to the right region and still hallucinate.



Question: "Is the dog wearing a collar?"		Ground Truth: Yes
Attention Metrics		Logit Lens Analysis
Spatial Entropy: $H_s = 0.321$ (Very low)		Peak layer: L14 ($\Delta\text{margin} = +9.57$)
Cluster Count: $C_k = 0$ (Concentrated)		Token "Yes" suppressed at L10–14
✗ Would predict: Reliable		✓ Correctly flags: Unreliable
Model Output: "No" Confidence: $P = 54.6\%$ INCORRECT		

Figure 4: **Case Study: High-Quality Attention, Wrong Answer (PaliGemma, Sample #31).** The image shows a dog on a surfboard clearly wearing a red collar. The model answers "No" despite exhibiting *excellent* attention: very low entropy ($H_s = 0.321$, bottom 15% of dataset) and concentrated focus ($C_k = 0$). Attention-based metrics would classify this as trustworthy. However, logit lens reveals the correct token "Yes" is suppressed at layer 14, correctly identifying unreliability.

6 Discussion

The results presented above fundamentally challenge the intuition that reliable multimodal generation is grounded in interpretable visual attention. Here, we analyze the implications of the "Cluster Failure" and the dominance of linguistic signals.

6.1 The Illusion of Grounding

The most striking finding of *VLM Reliability Probe (VRP)* is the near-zero correlation ($R = 0.001$) between "structural" attention and correctness. This suggests that the *Attention-Confidence Assumption* is an anthropomorphic fallacy. We assume that because humans look harder when they are trying to be precise, models must do the same.

However, the architecture of LLaVA (and similar VLMs) relies on a frozen CLIP encoder. The "looking" happens before the "reasoning." The visual features are extracted once and projected into the language space. If the visual encoder "misses" an object (e.g., fails to cluster on a baseball player), the LLM has no mechanism to "re-look." Conversely, even if the encoder perfectly clusters the object, the LLM can still hallucinate due to prior probabilities in its pre-training data (e.g., answering "2" because "2" is a common number of people in an image, regardless of the visual tokens).

To test whether this disconnect might be overcome through sophisticated feature engineering and machine learning, we conducted a supervised learning stress test (Section 4.3.6). Even when training ensemble classifiers with 11 attention-derived features (including cluster count, spatial entropy, concentration metrics, and polynomial interaction terms) and

full access to ground-truth labels, prediction accuracy remained at chance level (52–55%) across all three architectures. This definitive negative result confirms that attention patterns, regardless of how they are measured or combined, simply do not encode reliability information in current VLM architectures.

This indicates that **spatial attention maps are functional but uncalibrated**. They are necessary for the forward pass to work (as proven by our causal masking), but their internal entropy is not a proxy for the model’s subjective uncertainty.

6.2 Cross-Family Interpretation Beyond LLaVA

While LLaVA is the most deeply analyzed model, the same reliability taxonomy appears across all three families with model-specific signatures. **LLaVA-1.5** exhibits the strongest symbolic-detachment gap (early lock, late diffusion), which aligns with high probe separability in late layers. **PaliGemma-3B** integrates visual evidence earlier and more smoothly, yielding weaker late-layer separability and lower probe AUROC (0.738). **Qwen2-VL-7B** shows cyclical refinement and strong late-stage re-separation, consistent with high probe AUROC (0.971).

These differences suggest that reliability probing should be architecturally adaptive (e.g., layer selection and probe capacity per family), rather than assuming a one-size-fits-all late-layer template.

6.3 Reliability vs. Efficiency Trade-offs

While Self-Consistency (SC) is the gold standard for reliability ($R = 0.43$), it comes at a high computational cost: it requires $K = 10$ forward passes. For real-time applications (e.g., robotics), this is often prohibitive.

Our **Hidden State Probe** offers a compelling alternative:

- **Self-Consistency:** High Accuracy ($AUROC = 0.78$), High Cost ($10\times$ inference).
- **Learned Probe:** Moderate to High Accuracy ($AUROC = 0.96$ for LLaVA Combined), Zero Cost (overhead of a single linear layer).
- **Visual Metrics:** Low Accuracy ($AUROC = 0.50$), Low Cost.

The success of the Hidden State Probe confirms that the model’s reliability is encoded in the *linear subspace* of the final residual stream. This aligns with recent work in “Lie Detection” for LLMs, extending it to the multimodal domain. Future work should focus on distilling the signal from Self-Consistency into a single-pass value head, effectively training the model to predict its own consistency score.

7 Conclusion

As Multimodal Foundation Models become ubiquitous, the “black box” nature of their reliability poses a significant safety risk. In this work, we attempted to peer into that box using two lenses: the “eye” (visual attention) and the “voice” (linguistic consistency).

Our investigation yielded a definitive negative result for the “eye”: the spatial attention patterns of LLaVA-1.5 whether clustered, scattered, or focused are statistically orthogonal to the truthfulness of its answers. The “Cluster Failure” demonstrates that we cannot simply look at a heatmap to verify a model’s reasoning. Our layer-wise analysis elucidates the mechanism behind this disconnect, revealing a phenomenon of “Symbolic Detachment”: while models often exhibit “Early Locking” of visual features (e.g., at Layer 2), this focus frequently diffuses or stagnates by the final layers. Consequently, the attention mechanism, while causally necessary for feature extraction, does not encode the model’s subjective uncertainty about those features.

Instead, we must listen to the “voice.” The stability of the model’s generation process, measured via Self-Consistency, remains the single most effective proxy for truth ($R = 0.429$).

Furthermore, our internal probes reveal that this reliability signal is not emergent only at the output; it is encoded deeply in the hidden states of the late transformer layers (AUROC > 0.95).

We conclude that future research in trustworthy VLM design should shift focus from interpretability based on pixel-space grounding (which is often illusory) toward analyzing the latent dynamics of the language backbone, where the actual decision-making occurs.

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A Appendix

A.1 Extended Analysis: The Ensemble Attention Probe

In Section 6, we briefly introduced the “Ensemble Attention Probe” (Internal ID: Idea 4). Here we provide a detailed breakdown of its architecture and performance relative to other methods.

Motivation: The failure of unsupervised metrics (Cluster Count C_k) suggested that reliability is not encoded in simple geometric properties of the attention map (e.g., “is it sharp?”). However, we hypothesized that reliability might be encoded in *high-dimensional patterns* across multiple layers patterns too complex for human inspection but accessible to a non-linear classifier.

Architecture: We extracted the attention tensors $A^{(l)} \in \mathbb{R}^{H \times S}$ from all $L = 32$ layers of the Vicuna-7B backbone.

- **Input:** A concatenated vector of flattened attention maps from all layers:

$$x = \text{Concat}(\text{Vectorize}(A^{(1)}), \dots, \text{Vectorize}(A^{(32)})) \quad (11)$$

- **Model:** A 3-layer Multi-Layer Perceptron (MLP) with ReLU activations and Dropout ($p = 0.1$).
- **Dimensions:** Input $d_{in} = 32 \times 576 \rightarrow 1024 \rightarrow 512 \rightarrow 1$ (Binary Classification).

Results & Comparison: Table 4 details the performance of various probes. While the Ensemble Attention Probe significantly outperforms random chance and simple visual entropy, it is still inferior to Self-Consistency. This reinforces our main finding: *generation dynamics (consistency) are a stronger signal than internal state snapshots*.

Table 4: **Probe Performance Comparison.** The Supervised Ensemble (Idea 4) extracts some signal, but Consistency (Behavioral) remains superior.

Method	Type	AUROC	Cost (Inference)
Random Baseline	Statistical	0.500	1x
Focus Entropy (H_s)	Unsupervised Visual	0.504	1x
Cluster Count (C_k)	Unsupervised Visual	0.501	1x
Linear Probe (h_{last})	Supervised Ling.	0.620	1x
Ensemble Probe	Supervised Attn.	0.725	1x
Self-Consistency (SC)	Behavioral	0.784	10x

A.2 The Counting Anomaly: Severe Miscalibration

A critical discovery in our baseline testing was the model’s behavior on quantitative reasoning tasks. We refer to this as the “Counting Anomaly.”

The Phenomenon: On tasks asking “How many [objects] are in the image?”, LLaVA-1.5 exhibits **severe miscalibration**. As shown in our data, the model often assigns extremely high probability ($> 90\%$) to incorrect integers.

Case Study: Consider an image with 3 baseball players.

- **Ground Truth:** 3
- **Model Prediction:** “Four”
- **Token Confidence (P_{tok}):** 92% (Very High)
- **Visual Clusters (C_k):** 3 distinct clusters (Correctly perceives 3 objects).

This dissociation highlights a “Symbolic Detachment.” The visual encoder correctly identifies 3 regions (verified by $C_k = 3$), but the projection into the language space maps these features to the token “Four.” Because the language model is autoregressively coherent, it assigns high probability to the token “Four” despite it being factually grounded in “Three” visual features.

Conclusion: Token probability measures the model’s *fluency*, not its *grounding*. Self-Consistency mitigates this because, in the miscalibrated state, the model is likely to oscillate between “Four” and “Three” across different sampling temperatures, lowering the SC score.

A.3 Qualitative Failure Analysis

We analyzed specific instances where the “Attention-Confidence Assumption” broke down.

False Negatives (Good Attention, Bad Answer): In 15% of failure cases, the attention map was “perfect” (low entropy, high clustering on relevant objects). For example, in a “polling existence” task (POPE), the model attended solely to a chair while answering “No” to “Is there a chair?”. This suggests that the attention mechanism acted as a retrieval query that successfully found the feature, but the LLM decoder failed to interpret the retrieved feature as “existence.”

False Positives (Bad Attention, Good Answer): In 22% of correct cases, the model exhibited “scattered” attention (high entropy, $H_s > 4.5$). This frequently occurred in background scene questions (e.g., “Is this a rainy day?”). The model likely relied on global texture features pooled from the entire image rather than specific object attention, yet standard interpretability metrics would penalize this as “unfocused.”

A.4 Cross-Model Experiment Details

A.5 Family-Specific Reliability Patterns

To complement the LLaVA-centric deep dive, we summarize family-specific behaviors: **LLaVA-1.5** shows a long early-lock plateau followed by late diffusion and strong final-layer probe signal; **PaliGemma-3B** shows earlier integration and weaker late-layer margin separation; **Qwen2-VL-7B** shows iterative re-integration cycles with strong late reliability separation. This appendix section is intended to balance interpretive depth across all three families under page constraints in the main paper.

We conducted extensive experiments across three VLM architectures to validate generality.

Model Architectures:

- **LLaVA-1.5-7B:** 32 transformer layers, 32 attention heads per layer. Uses frozen CLIP ViT-L/14 visual encoder with Vicuna-7B language backbone. Visual tokens projected via 2-layer MLP.
- **PaliGemma-3B** (Google): 18 transformer layers, 8 attention heads per layer. Uses SigLIP visual encoder with Gemma language backbone. Visual tokens projected via linear layer.
- **Qwen2-VL-7B-Instruct** (Alibaba): 28 transformer layers with Grouped Query Attention (28 heads, 4 KV heads). Native multimodal architecture with interleaved visual tokens and dynamic resolution support.

A.6 LLaVA Detailed Analysis

Key Insight: Correctness emerges *before* final answer selection. Margin trajectories diverge at Layer 21 and peak at Layer 24, suggesting reliability is determined in mid-layers, not at the final output. Table 5 presents the complete LLaVA analysis.

Table 5: **LLaVA-1.5-7B Complete Analysis.** Layer-wise computational pipeline, neuron-level findings, and causal validation.

<i>Layer-wise Computational Pipeline</i>			
Layers	Role	Δmargin	Dominant Component
0–16	Feature extraction	Low variance	—
17	Early prediction	—	82.3% probe accuracy
19	Early boosting	+0.53	MLP
21–28	Suppression	−0.85 to −2.27	Attention (72%)
24	Max separation	—	Largest correct/incorrect gap
29	Neuron commitment	—	86.3% probe, 5.7% sparse
30	Answer boosting	+2.61	MLP
31	Final decision	+8.80	MLP (72%)
<i>Key Neurons (Layer 31)</i>			
Neuron ID	Type	$\Delta\text{activation}$	Functional Role
1512	Success	+27.23	Answer confidence
1360	Failure	−3.11	Failure detection
3839	Failure	−3.08	Failure detection
2660	Failure	−2.95	Failure detection

A.7 Implementation and Hardware Details

Hardware: All experiments were conducted on a compute cluster provided by RunPod and Lambda Labs.

- **GPU:** NVIDIA A100 (80GB VRAM)
- **CPU:** AMD EPYC 7742 64-Core
- **RAM:** 512 GB System RAM

Software Environment:

- **Framework:** PyTorch 2.1.0 with CUDA 12.1
- **Model Source:** HuggingFace transformers library (llava-hf/llava-1.5-7b-hf).
- **Attention Extraction:** We utilized PyTorch hooks (`register_forward_hook`) on the MultiheadAttention modules of the Vicuna decoder layers (specifically layers 16–32, where visual-linguistic integration is hypothesized to be densest).

A.8 Limitations and Future Work

Model Scale: Our study was limited to LLaVA-1.5 (7B). It is possible that larger models (e.g., LLaVA-34B or GPT-4V) exhibit stronger alignment between attention and truthfulness due to better reinforcement learning from human feedback (RLHF).

Computational Cost: The most reliable metric found, Self-Consistency, requires $K = 10$ inference passes. This is prohibitively expensive for low-latency edge applications.

Future Direction: We propose that future work should focus on *distillation*. Since Self-Consistency provides a high-quality “silver label” for reliability ($R = 0.43$), we can curate a dataset of (Image, Question, Answer, SC-Score) and fine-tune a value head on top of the VLM to predict the SC-Score in a single pass. This would combine the accuracy of consistency with the efficiency of a probe.